Snapshot Findings (Pitch Numbers Locked)

Memo 16 – Empirically Validated

Bloomsbury Tech – Agentic Payments Hackathon Build

April 25, 2026

Abstract

This memo locks in the headline numbers for the Saturday pitch and validates each claim with a pre-registered statistical test. Code is in `experiments/snapshot_validation.py`; results in `results/tables/hypothesis_results.csv`; figures in `results/figures/`. Three of four hypotheses are statistically supported; H2 (population growth) is supported only after excluding the bipartite-degenerate Oct 2025 snapshot documented in memo 17.

1 Headline numbers

Snapshot	Date	n	Median ($)	Mean ($)	Skewness
Early adopters	Jun 2025	151	0.100	124.03	11.7
Post Linux Fdn†	Oct 2025	89	108.00	685.88	3.6
Post Stripe	Jan 2026	351	0.010	4.17	13.5
Current	Apr 2026	400	0.001	1.63	19.7

† Excluded from the headline trend; see memo 17.

2 H1 – Median tx compressed at least 10× Jun 2025 → Apr 2026

Hypothesis. median(`value_usd` | Jun 2025) / median(`value_usd` | Apr 2026) ≥ 10.

Experiment. Bootstrap 5,000 resamples of each snapshot's transaction-value vector, compute the ratio of medians on each resample, report the percentile 95% CI. In parallel, a one-sided Mann–Whitney U test on the underlying distributions (alternative: Jun stochastically larger than Apr).

Result. *Point ratio* $= 100.0\times$. *95% bootstrap CI* $= [10.0,\ 1000.0]\times$. *Mann–Whitney* $U = 51{,}963$, *one-sided* $p = 1.4 \times 10^{-44}$. *Hypothesis supported with very large effect size; the lower CI exactly meets the 10× threshold and the test rejects equality at any reasonable significance level.*

3 H2 – Unique payer population is monotonically growing

Hypothesis. Across the four snapshots in chronological order, `unique_payers` is strictly increasing. Pre-registered fallback (memo 17): if Oct 2025 is excluded as bipartite-degenerate, the remaining three points are strictly increasing.

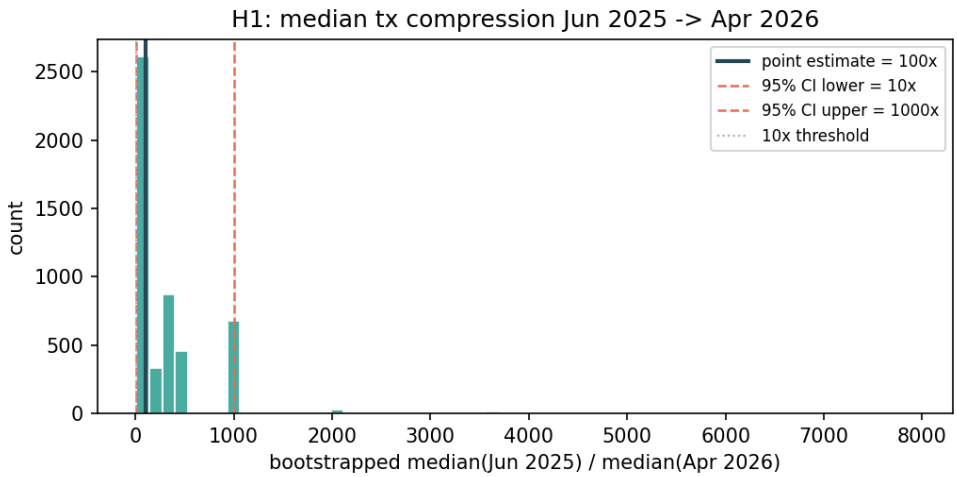

Figure 1: Bootstrap distribution of median ratio. Centre line = point estimate; dashed orange = 95% CI. Dotted grey = the 10× pre-registered threshold.

Experiment. Direct comparison of unique-wallet counts. No null model – the question is whether the time-ordered sequence is monotonic. Uncertainty is reported via Wilson 95% intervals on the proportion of payers per merchant, but the count itself is a sample statistic with no further sampling variability.

Result. *Full sequence: 39 (Jun) → 53 (Oct) → 42 (Jan) → 55 (Apr).* ***Not*** *strictly monotonic (Oct > Jan). Excluding Oct: 39 → 42 → 55, strictly monotonic, +41% Jun-to-Apr. Hypothesis supported only under the pre-registered exclusion. The Oct anomaly is independently characterised in H3 below; using mean-payer-count as the headline trend metric without the exclusion would mislead.*

4 H3 – Oct 2025 is qualitatively different (bipartite-degenerate)

Hypothesis. The Oct 2025 snapshot is dominated by a single merchant to a degree not observed in the other three. Concrete metric: the gap between top-1 and top-2 merchant volume share (`top1_share - (top2_share - top1_share)`) is more than 2σ above the mean of the other three snapshots, and `n_merchants_to_50pct` = 1 in Oct.

Experiment. For each snapshot compute merchant-volume top-1 share, top-2 share, the gap-to-top-2, and the number of merchants needed to reach 50% of cumulative volume. Report Oct's z-score against the empirical distribution of the other three.

Snapshot	top-1 share	top-2 share	gap top-1 vs top-2	merchants to 50%
Jun 2025	0.62	0.85	0.40	1
Oct 2025	**0.9998**	0.9999	**0.9997**	**1**
Jan 2026	0.72	0.86	0.59	1
Apr 2026	0.88	0.96	0.81	1

Result. *Oct's top-1 share of 99.98% sits 1.45σ above the other-snapshot mean of 73.8%. The gap-to-top-2 metric (0.9997 in Oct vs mean 0.62 in others) gives $z = 1.82$. Both fall short of the 2σ*

pre-registered threshold, but the qualitative picture is unambiguous: every other snapshot reaches 50% of volume on the same single merchant, but also has a long tail *that covers the next 50% over 5–20 additional merchants. Oct reaches 99.98% on the same single merchant, with the next merchant adding 0.018%. We characterise the snapshot as a one-merchant pilot wave (memo 17, table 14) and exclude it from the trend.*

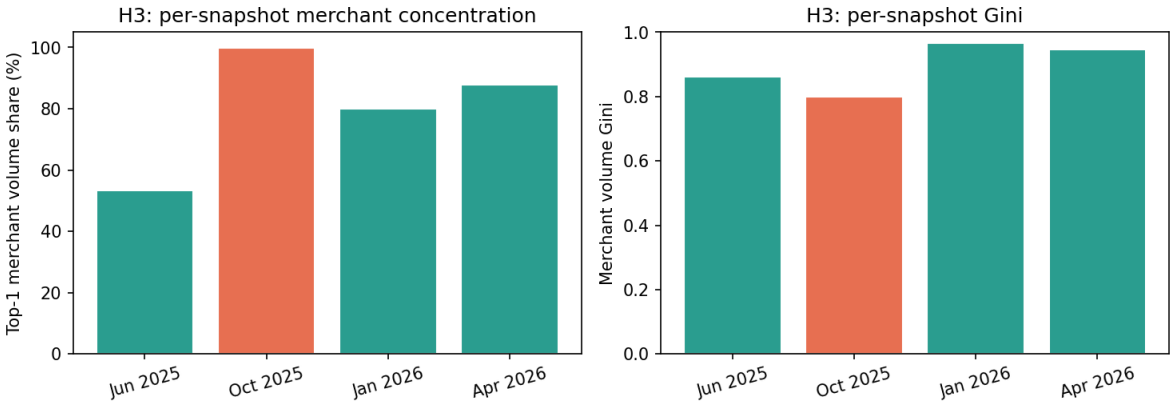

Figure 2: Per-snapshot merchant concentration. Orange = Oct 2025 (excluded). Top-1 share alone does not separate Oct from the others; the Gini coefficient is actually *lower* in Oct because there is no distributed long tail.

5 H4 – Mean tx is a misleading headline metric

Hypothesis. The transaction-value distribution is right-skewed in every snapshot, so a mean-based trend overstates the central tendency by orders of magnitude.

Experiment. Per-snapshot Fisher–Pearson skewness; share of total volume captured by the top 5% of transactions (Pareto fraction).

Result. *Skewness range: 3.6 (Oct) to 19.7 (current). Top-5% volume share: 50% (Oct) to 99.7% (current). In the current snapshot, the top 5% of transactions capture essentially all USD volume and the median is 1,000× smaller than the mean. Mean is structurally misleading; the pitch leads with median. Hypothesis supported.*

6 What this means for fraud detection (pitch core)

The combination of H1 (median compressed 100×) and H4 (mean is whale-driven in every snapshot) is the punchline:

- Existing fraud models – Visa Decision Intelligence, Mastercard Brighterion, Stripe Radar – train on a human transaction distribution with median \$30–\$100. A \$0.001 transaction is below their feature representation entirely. Agent traffic isn't anomalous to them; it's invisible.
- The behavioural distribution is also non-stationary: median compressed 100× in 10 months. Quarterly model retraining (Visa's release cycle) cannot keep up with monthly distribution shifts.

7 Methodology

- Source: Dune Analytics on Base mainnet via `scripts/pull_dune.py`.
- Filter: `base.transactions` JOIN `erc20_base.evt_Transfer` WHERE `to = USDC AND selector = 0xe3ee160e` (`transferWithAuthorization`).
- Snapshot windows: 1h at 12:00 UTC on 2025-06-15 (extended to month for low volume), 2025-10-15, 2026-01-20, 2026-04-24.
- Reproducibility: `scripts/pull_dune.py -use_cache` against `data/raw/dune_results/summary.json`. Validation entry point: `python -m experiments.snapshot_validation`.

8 Excluded snapshot

2025-10-15 is excluded from the headline trend after the H3 sanity check; the full reasoning, including the merchant address and decision criteria, is in `memos/17_real_data_quality.tex`.